

NCC VOLUME TWO, ENERGY EFFICIENCY

Case Study – Typical Masonry Veneer House

1. INTRODUCTION

The energy efficiency elemental Deemed-to-Satisfy (DtS) Provisions in Volume Two of the NCC for Class 1 and 10 buildings are relatively detailed because they cover all of the major components of a house and because they vary considerably depending on the applicable climate zone. Given this level of complexity, it is important that practitioners have a good understanding of how to apply the energy efficiency provisions to a house.

The purpose of this case study is to examine how the energy efficiency elemental DtS Provisions apply to a typical masonry veneer dwelling. A fairly typical house has been chosen to provide a good understanding of how the provisions are applied.

Of course, houses of this type can also be designed using house energy rating software in accordance with the DtS pathway. However, this study will focus on the elemental DtS Provisions.

2. THE PROJECT

A three bedroom masonry veneer house is to be constructed in climate zone 6. It has cavity masonry external walls in the lounge room, and the wall between the house and garage is a stud wall. The house is to be constructed on a concrete slab-on-ground and will have a conventional pitched tiled roof with a roof upper surface solar absorptance value of more than 0.6 and a flat ceiling.

The house has 3 ceiling fans and there are 4 downlights penetrating the ceiling.

An electric in-slab heating system will be installed under the bathroom. It will have a thermostat and time switch and the power load is 120 W/m².

A ducted reverse-cycle air-conditioning system will be installed throughout the rest of the house (not the garage).

The lighting system totals 500 Watts for the house and 60 Watts for the garage. There are no lighting controls.

A floor plan, elevations and section of the subject house are shown in the next few pages.

FIGURE 1 - FLOOR PLAN

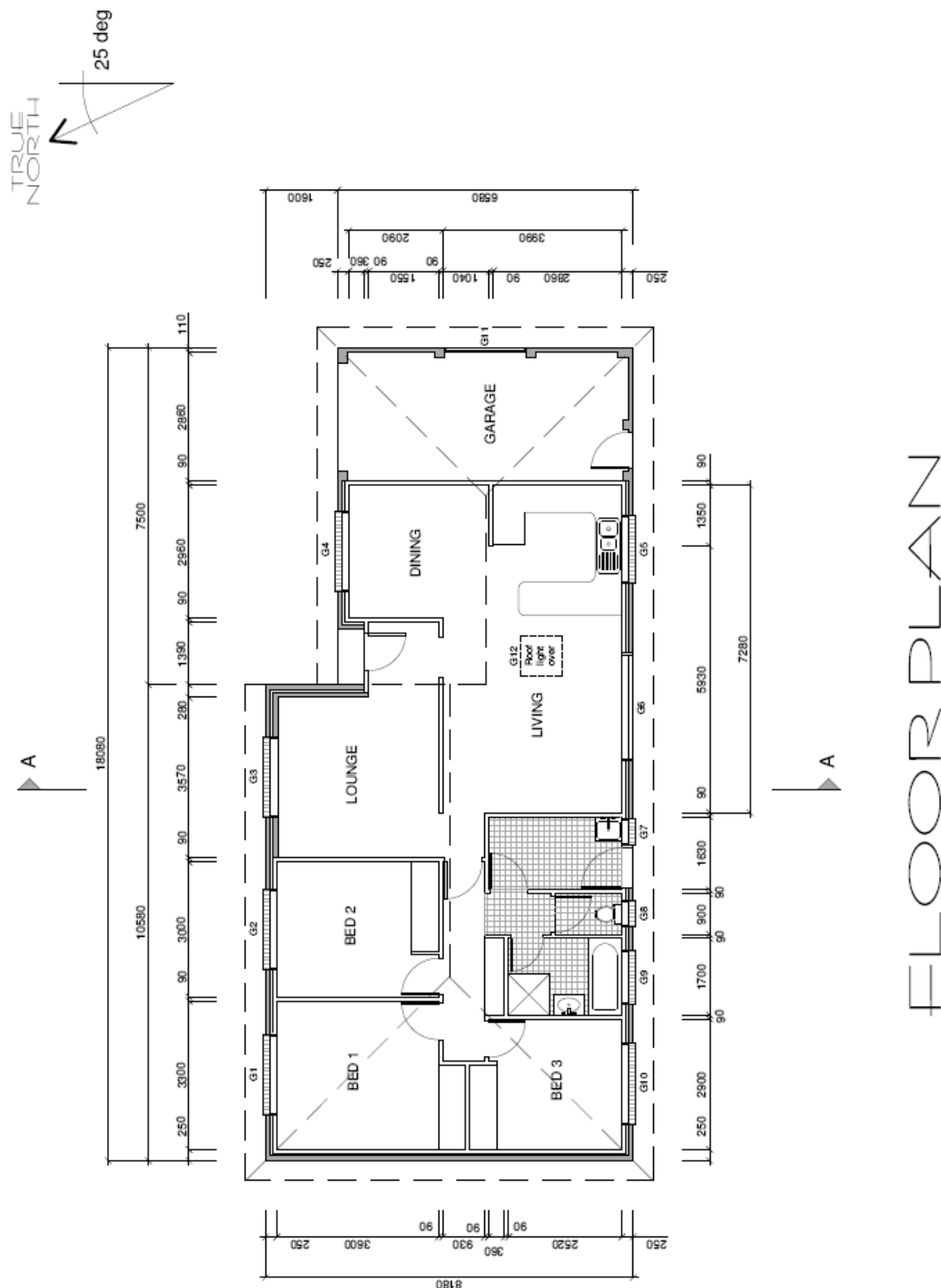


FIGURE 2 - ELEVATIONS

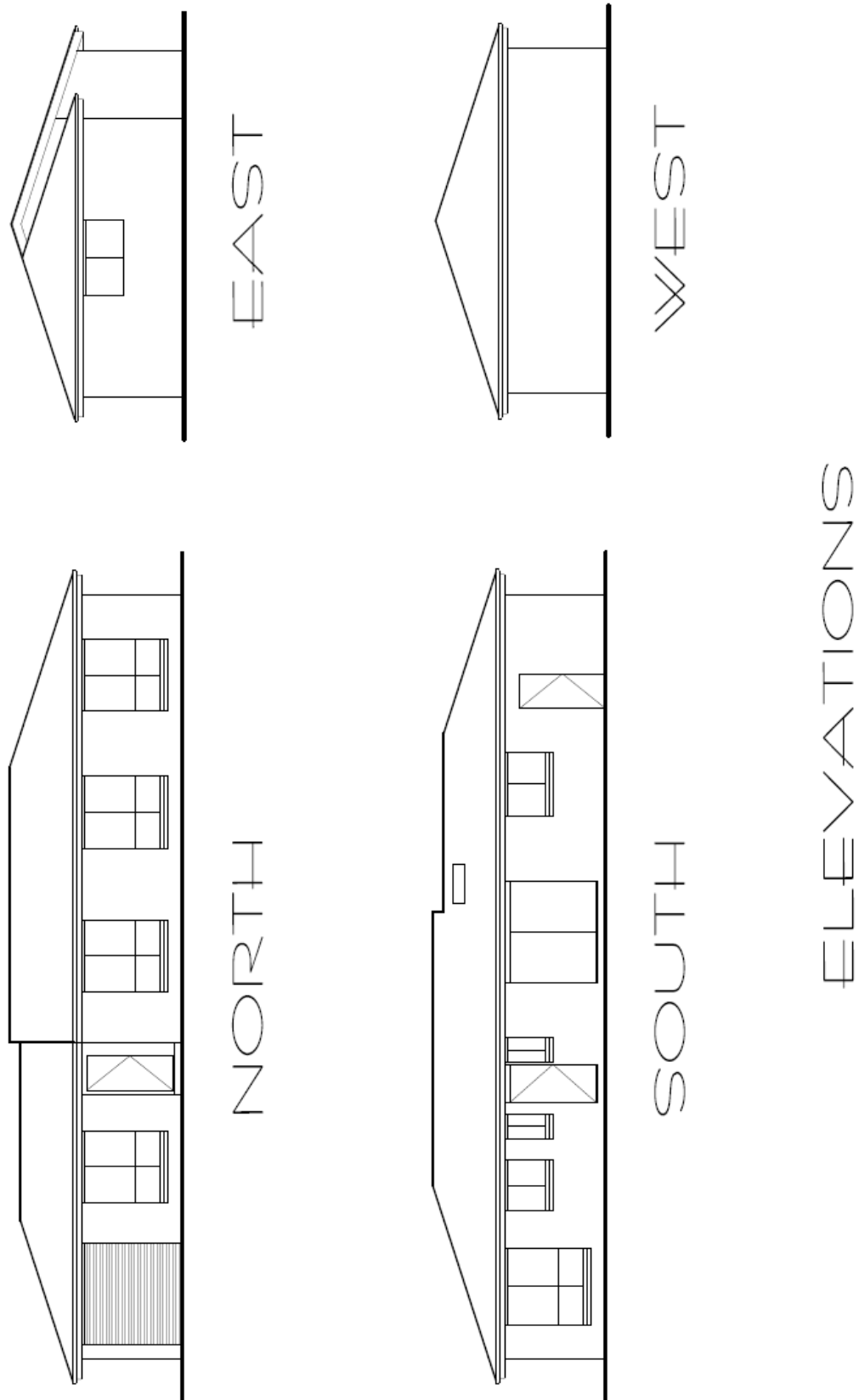
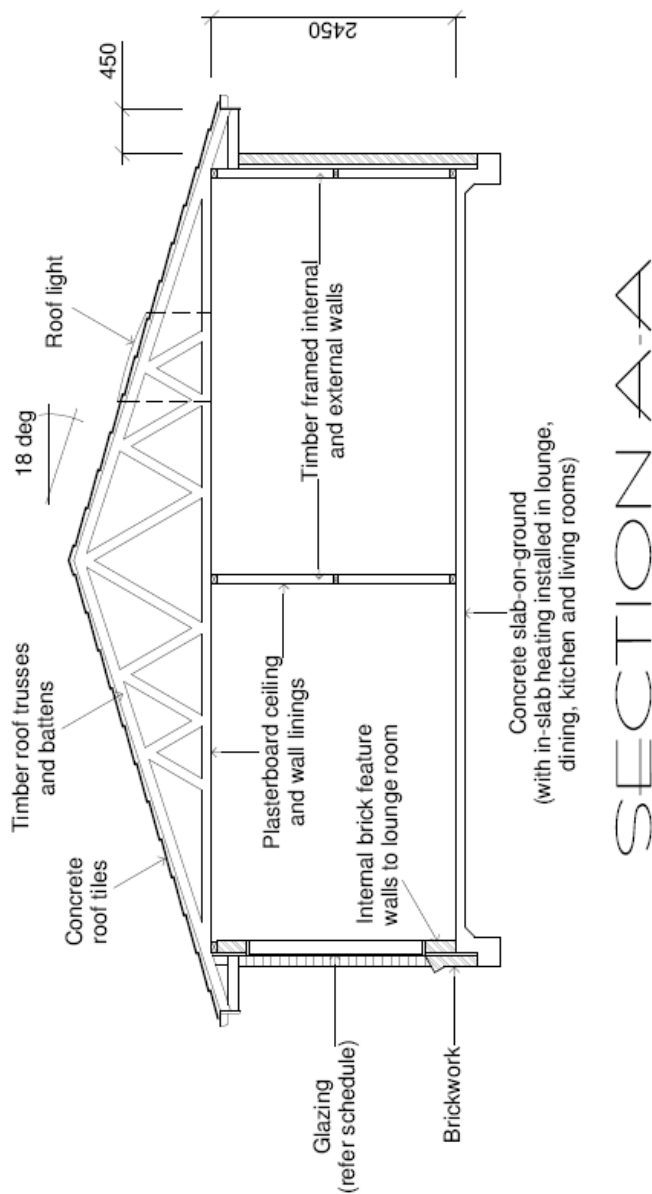


FIGURE 3 – SECTION AA AND GLAZING SCHEDULE



GLAZING SCHEDULE			
G1	1800h x 1800w Sliding Window	G7	900h x 600w Sliding Window
G2	1800h x 1800w Sliding Window	G8	900h x 600w Sliding Window
G3	1800h x 1800w Sliding Window	G9	900h x 1200w Sliding Window
G4	1800h x 1800w Sliding Window	G10	1800h x 1800w Sliding Window
G5	900h x 1500w Sliding Window	G11	900h x 1800w Sliding Window
G6	2100h x 2400w Sliding Door	G12	900 x 900 Roof Light

3. CLIMATE ZONE

The first fundamental matter that needs to be determined before applying the DtS Provisions is the climate zone in which the house is to be located. As "climate zone" is a defined term, an explanation of this term is contained within Part 1.1.1 of Volume Two. There is also a map of Australia showing the extent of each zone and a table detailing the applicable climate zone for common locations.

For locations that are more difficult to determine, a suite of State and Territory climate zone maps can be viewed on the ABCB website, www.abcb.gov.au.

In this case, the applicable climate zone is 6.

4. BUILDING FABRIC

Having determined the applicable climate zone, the next step is to examine Part 3.12.1 and determine what thermal performance is required for the fabric of the house.

4.1 Roof

Roofs with a roof upper surface solar absorptance value more than 0.6 in climate zone 6 are required to achieve a minimum Total R-Value of 5.1 in the upwards direction (refer Table 3.12.1.1a). According to Figure 3.12.1.1, a ventilated pitched tiled roof with a flat ceiling achieves a Total R-Value of 0.23 (upward). This means that additional insulation of minimum R-Value 4.87 ($5.1 - 0.23$) in the upward direction is needed. This can be achieved by installing bulk insulation or a combination of bulk and reflective insulation. There are only three ceiling fans and 4 downlights penetrating the ceiling insulation. Assuming a 200 mm clearance for each downlight, (i.e. 0.04m^2 per downlight) no compensation for penetrations is needed as the total penetrations are less than 0.5% of the floor area.

4.2 Roof lights

The roof light serving the living, kitchen and dining rooms has a roof shaft height of 750 mm and constitutes 2.3% of the floor area of the space served so is therefore required to comply with Table 3.12.1.2. The roof light shaft index is approximately 0.83 ($750 \text{ height} / 900 \text{ opening}$), which means that the roof light is required to achieve a Total System U-Value of not more than 5.7 and a Total System SHGC of not more than 0.72. Using the "worst case" performance values in the Explanatory Information following Table 3.12.1.2, a flat framed panel with a double layer clear element, but without a ceiling diffuser achieves compliance.

4.3 External walls

Most of the external walls have a surface density of less than 220 kg/m^2 so are required to comply with Table 3.12.1.3a, which requires a minimum Total R-Value of 2.8 to be achieved. Using the information in Figure 3.12.1.3, this means that additional insulation with a minimum R-Value of 2.24 ($2.8 - 0.56$) is required within the masonry veneer external walls. Note that the air space must be maintained so the frame size will need to be consistent with the insulation thickness.

It is also worth noting, however, that if the external walls had a surface density more than 220 kg/m² and the lowest storey containing habitable rooms had a concrete slab-on-ground floor or masonry internal walls, under Table 3.12.1.3b it would only be necessary to incorporate insulation with an R-Value not less than 1.0 (refer option (c) for climate zone 6). Cavity masonry walls are considered to achieve a surface density of more than 220 kg/m² which means that the cavity masonry walls to the lounge room only require insulation with an R-Value not less than 1.0.

The external walls of the garage are not affected by these provisions because the garage is not a conditioned space (refer Part 3.12.1).

4.4 Floors

The provisions do not require the insulation of concrete slab-on-ground floors except where they contain an in-slab heating or cooling system or are located in climate zone 8 (refer 3.12.1.5(c)(i) and (ii)). Given that it is proposed to install an in-slab heating system within the bathroom, insulation with an R-Value not less than 1.0 must be installed around the external vertical edge of the slab as described in 3.12.1.5(c).

4.5 Attached Class 10a buildings

In climate zone 6, the provisions require that either the external walls of attached garages or the internal wall separating the attached garage from the house achieve the same level of thermal performance as the external walls of the house. In this instance, because the external walls of the garage are of single skin masonry, and the internal wall framed, it would be easier for the separating wall to achieve the required level of thermal performance.

As previously discussed, the external walls of the house are required to achieve a minimum Total R-Value of 2.8. This can be achieved by the external garage wall or by the internal separating wall by the addition of bulk insulation. However, it should be noted that the inherent R-Value of the separating wall can be calculated to be 0.45 (an outdoor air film, a layer of plasterboard, a 90 mm stud frame with airspace, a layer of plasterboard and an indoor air film) using the information in Specification J1.5 of Volume One. This means that the R-Value of the additional insulation can be reduced to 2.35 (2.8 – 0.45) which could be achieved by incorporating bulk insulation with an R-Value of 2.5 if the airspace is maintained. If the airspace is filled, the Total R-Value should be reduced by 0.17 to take account of the loss of airspace.

5. GLAZING

Part 3.12.2 details the calculation to be used to determine compliance of glazing with the energy efficiency requirements. To simplify this process, the ABCB has developed a glazing calculator, which can be downloaded from the ABCB's website www.abcb.gov.au. The calculator is a spreadsheet that automatically determines whether the glazing performance targets have been achieved based on the input of the glazing characteristics.

In terms of the case study house, the results from using the glazing calculator are shown below. Note that window G11 is not included as the garage is not a conditioned space.

The P and H values for each window are as follows:

P= 0.6 m (450 mm eave plus 150 mm gutter), H= Height of the window plus 100 mm

These results are based on the use of the “worst case” performance values in the Explanatory Information following 3.12.2.1 for single clear glass with aluminium frames. The results show that the solution fails on conductance which is 119% more than the allowance. The conductance fail can be due to the Total System U-Value being too high or Total System Solar Heat Gain Coefficient being too low.

FIGURE 4 - GLAZING CALCULATOR WITH SINGLE GLAZING - FAIL

NCC VOLUME TWO GLAZING CALCULATOR (first issued with NCC 2014) HELP

Building name/description: **Case Study - Typical Brick Veneer House** Climate zone: **6**

Storey: **1** Floor Construction: **Direct contact** Area: **116m²** Wall insulation option chosen for 3.12.1.4: **No wall insulation concession used**

Air Movement: **Standard** Area of storey: **116m²** Area of glazing: **24.8m² (21% of area of storey)**

Number of rows for table below: **10 (as currently displayed)**

GLAZING ELEMENTS, ORIENTATION SECTOR, SIZE and PERFORMANCE CHARACTERISTICS						SHADING		CALCULATION DATA				CALCULATED OUTCOMES					
Glazing element		Orientation		Size		Performance		P&H or device		Exposure		Size		Conductance - FAILED		Solar heat gain - PASSED	
ID	Description (optional)	Facing sector	Height (m)	Width (m)	Area (m²)	Total System U-Value (AFRC)	Total System SHGC (AFRC)	P (m)	H (m)	P/H	Es	Area used (m²)	U x area / winter access	Element share of % of allowance used	SHGC x Es x area	Element share of % of allowance used	
1	G1	NE	1.80	1.80		7.90	0.81	0.60	1.90	0.32	0.66	3.24	1.83	13% of 219%	1.7	13% of 77%	
2	G2	NE	1.80	1.80		7.90	0.81	0.60	1.90	0.32	0.66	3.24	1.83	13% of 219%	1.7	13% of 77%	
3	G3	NE	1.80	1.80		7.90	0.81	0.60	1.90	0.32	0.66	3.24	1.83	13% of 219%	1.7	13% of 77%	
4	G4	NE	1.80	1.80		7.90	0.81	0.60	1.90	0.32	0.66	3.24	1.83	13% of 219%	1.7	13% of 77%	
5	G5	SW	0.90	1.50		7.90	0.81	0.60	1.00	0.60	0.58	1.35	0.76	5% of 219%	0.6	5% of 77%	
6	G6	SW	2.10	2.40		7.90	0.81	0.60	2.20	0.27	0.77	5.04	2.85	20% of 219%	3.1	23% of 77%	
7	G7	SW	0.90	0.60		7.90	0.81	0.60	1.00	0.60	0.58	0.54	0.31	2% of 219%	0.3	2% of 77%	
8	G8	SW	0.90	0.60		7.90	0.81	0.60	1.00	0.60	0.58	0.54	0.31	2% of 219%	0.3	2% of 77%	
9	G9	SW	0.90	1.20		7.90	0.81	0.60	1.00	0.60	0.58	1.08	0.61	4% of 219%	0.5	4% of 77%	
10	G10	SW	1.80	1.80		7.90	0.81	0.60	1.90	0.32	0.73	3.24	1.83	13% of 219%	1.9	14% of 77%	

IMPORTANT NOTICE AND DISCLAIMER IN RESPECT OF THE GLAZING CALCULATOR
The Glazing Calculator has been developed by the ABCB to assist in developing a better understanding of glazing energy efficiency parameters. While the ABCB believes that the Glazing Calculator, if used correctly, will produce accurate results, it is provided "as is" and without any representation or warranty of any kind, including that it is fit for any purpose or of merchantable quality, or functions as intended or at all.

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In order to achieve compliance, the performance values were replaced with values for clear double glazing with timber framing, as shown in the next figure.

FIGURE 5 - GLAZING CALCULATOR WITH DOUBLE GLAZING – FAIL

NCC VOLUME TWO GLAZING CALCULATOR (first issued with NCC 2014)

Building name/description: **Case Study - Typical Brick Veneer House** Climate zone: **6**

Storey: **1** Floor Construction: **Direct contact** Area: **116m²** Wall insulation option chosen for 3.12.1.4

Air Movement: **Standard** Area of storey: **116m²** **No wall insulation concession used**

Area of glazing: **24.8m² (21% of area of storey)**

Number of rows for table below: **10 (as currently displayed)**

GLAZING ELEMENTS, ORIENTATION SECTOR, SIZE and PERFORMANCE CHARACTERISTICS					SHADING		CALCULATION DATA		CALCULATED OUTCOMES							
Glazing element	Orientation	Size			Performance		P&H or device		Exposure	Size	Conductance - FAILED		Solar heat gain - PASSED			
ID	Description (optional)	Facing sector	Height (m)	Width (m)	Area (m²)	Total System U-Value (AFRC)	Total System SHGC (AFRC)	P (m)	H (m)	P/H	Es	Area used (m²)	U x area / winter access	Element share of % of allowance used	SHGC x Es x area	Element share of % of allowance used
1 G1		NE	1.80	1.80		3.20	0.68	0.60	1.90	0.32	0.66	3.24	0.88	13% of 106%	1.5	13% of 65%
2 G2		NE	1.80	1.80		3.20	0.68	0.60	1.90	0.32	0.66	3.24	0.88	13% of 106%	1.5	13% of 65%
3 G3		NE	1.80	1.80		3.20	0.68	0.60	1.90	0.32	0.66	3.24	0.88	13% of 106%	1.5	13% of 65%
4 G4		NE	1.80	1.80		3.20	0.68	0.60	1.90	0.32	0.66	3.24	0.88	13% of 106%	1.5	13% of 65%
5 G5		SW	0.90	1.50		3.20	0.68	0.60	1.00	0.60	0.58	1.35	0.37	5% of 106%	0.5	5% of 65%
6 G6		SW	2.10	2.40		3.20	0.68	0.60	2.20	0.27	0.77	5.04	1.38	20% of 106%	2.6	23% of 65%
7 G7		SW	0.90	0.60		3.20	0.68	0.60	1.00	0.60	0.58	0.54	0.15	2% of 106%	0.2	2% of 65%
8 G8		SW	0.90	0.60		3.20	0.68	0.60	1.00	0.60	0.58	0.54	0.15	2% of 106%	0.2	2% of 65%
9 G9		SW	0.90	1.20		3.20	0.68	0.60	1.00	0.60	0.58	1.08	0.29	4% of 106%	0.4	4% of 65%
10 G10		SW	1.80	1.80		3.20	0.68	0.60	1.90	0.32	0.73	3.24	0.88	13% of 106%	1.6	14% of 65%

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It can be seen that the solution still fails on conductance but now only by 6% over the allowance. As a further refinement, the 600 mm eave on the north-east side has been reduced to 400 mm. This refinement achieves compliance because the problem was with winter conductance, so more north-east sun is beneficial and improves the performance.

FIGURE 6 - GLAZING CALCULATOR - PASS

NCC VOLUME TWO GLAZING CALCULATOR (first issued with NCC 2014)

Building name/description: **Case Study - Typical Brick Veneer House** Climate zone: **6**

Storey: **1** Floor Construction: **Direct contact** Area: **116m²** Wall insulation option chosen for 3.12.1.4

Air Movement: **Standard** Area of storey: **116m²** **No wall insulation concession used**

Area of glazing: **24.8m² (21% of area of storey)**

Number of rows for table below: **10 (as currently displayed)**

GLAZING ELEMENTS, ORIENTATION SECTOR, SIZE and PERFORMANCE CHARACTERISTICS					SHADING		CALCULATION DATA		CALCULATED OUTCOMES - OK (if inputs are valid)							
Glazing element	Orientation	Size			Performance		P&H or device		Exposure	Size	Conductance - PASSED		Solar heat gain - PASSED			
ID	Description (optional)	Facing sector	Height (m)	Width (m)	Area (m²)	Total System U-Value (AFRC)	Total System SHGC (AFRC)	P (m)	H (m)	P/H	Es	Area used (m²)	U x area / winter access	Element share of % of allowance used	SHGC x Es x area	Element share of % of allowance used
1 G1		NE	1.80	1.80		3.20	0.68	0.40	1.90	0.21	0.76	3.24	0.83	13% of 99%	1.7	14% of 70%
2 G2		NE	1.80	1.80		3.20	0.68	0.40	1.90	0.21	0.76	3.24	0.83	13% of 99%	1.7	14% of 70%
3 G3		NE	1.80	1.80		3.20	0.68	0.40	1.90	0.21	0.76	3.24	0.83	13% of 99%	1.7	14% of 70%
4 G4		NE	1.80	1.80		3.20	0.68	0.40	1.90	0.21	0.76	3.24	0.83	13% of 99%	1.7	14% of 70%
5 G5		SW	0.90	1.50		3.20	0.68	0.60	1.00	0.60	0.58	1.35	0.35	5% of 99%	0.5	4% of 70%
6 G6		SW	2.10	2.40		3.20	0.68	0.60	2.20	0.27	0.77	5.04	1.29	20% of 99%	2.6	21% of 70%
7 G7		SW	0.90	0.60		3.20	0.68	0.60	1.00	0.60	0.58	0.54	0.14	2% of 99%	0.2	2% of 70%
8 G8		SW	0.90	0.60		3.20	0.68	0.60	1.00	0.60	0.58	0.54	0.14	2% of 99%	0.2	2% of 70%
9 G9		SW	0.90	1.20		3.20	0.68	0.60	1.00	0.60	0.58	1.08	0.28	4% of 99%	0.4	3% of 70%
10 G10		SW	1.80	1.80		3.20	0.68	0.60	1.90	0.32	0.73	3.24	0.83	13% of 99%	1.6	13% of 70%

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6. BUILDING SEALING

The next step is to examine Part 3.12.3 to determine whether the subject house is required to be sealed in order to control unwanted air leakage.

Note that the garage is not affected by these provisions because it is used only for the accommodation of vehicles (refer 3.12.3(b)(iii)). This concession is in place to avoid a build up of carbon monoxide if a car is running in a closed and sealed garage.

6.1 Chimneys and flues

There are no chimneys and flues installed in the case study house, therefore Part 3.12.3.1 is not applicable.

6.2 Roof lights

In the case study house, the living, kitchen and dining rooms are served by a roof light. In climate zone 6, roof lights serving habitable rooms must be sealed or capable of being sealed irrespective of whether the rooms are conditioned or not, therefore this roof light must comply with Part 3.12.3.2. The roof light is not constructed with a diffuser or a shutter system so to achieve the required performance, the roof light is constructed with a weatherproof seal.

6.3 External windows and doors

External windows and doors in climate zone 6 serving a conditioned space or habitable room must be sealed at each edge to restrict unwanted air leakage. In this case, the garage is the only space in the building that is not artificially heated or cooled (conditioned) thus the external window and doors serving the garage will not require sealing.

External windows and doors may be fitted with a foam or rubber compressible strip, fibrous seal or the like to achieve compliance. However the bottom edge of an external swing door must be fitted with a draft protection device or comply with AS 2047.

6.4 Exhaust fans

There are no exhaust fans installed in the house therefore Part 3.12.3.4 is not applicable.

6.5 Construction of roofs, walls and floors

In climate zone 6, roofs, external walls, external floors and any openings forming part of the external fabric of the house must comply with Part 3.12.3.5 if serving a habitable room irrespective of whether conditioned or not. Non-habitable rooms that are conditioned are also required to comply. This means that the garage's external fabric and its openings are exempted from complying with 3.12.3.5(b).

Compliance with this part can be achieved by caulking, skirting, architraves and cornices.

6.6 Evaporative coolers

There are no evaporative coolers installed in the house therefore Part 3.12.3.6 does not apply.

7. AIR MOVEMENT

Part 3.12.4 requires compliance with Part 3.8.5 for climate zone 6 so Part 3.12.4 has no additional requirements.

8. SERVICES

The last step is to examine Part 3.12.5 to establish provisions applicable to the house's services to minimise the amount of energy lost through air-conditioning ductwork, central heating water piping, lighting, swimming pool or spa pool heating and pumping. These provisions apply to Class 1 and 10a buildings and Class 10b swimming pools.

8.1 Insulation of services

The provisions require thermal insulation complying with AS/NZS 4859.1 for central heating water piping and heating and cooling ductwork to be protected against the adverse effects of weather and sunlight and to be able to withstand the temperatures within the piping or ductwork.

8.2 Central heating water piping

There is no central heating water piping, except for the in-slab heating system for the bathroom. This piping is exempted from being insulated as stated in Table 3.12.5.1 (1)(b) as the insulation would impede the heating of the slab.

8.3 Heating and cooling ductwork

In climate zone 6, the ductwork of the reverse-cycle air-conditioning system is required to achieve a minimum material R-Value of 1.5. This value can be reduced by 0.5 if the ducts are in a roof space that has insulation of not less than R0.5 directly beneath the roofing or under a suspended floor with an enclosed perimeter, as described in Table 3.12.5.2. Additionally, the fittings of the reverse-cycle air-conditioning system must achieve a minimum material R-Value of 0.4.

Note that 3.12.5.3(c) exempts ductwork inside the conditioned space.

The ductwork and fittings are required to be sealed against air loss. Rigid ductwork must be sealed by closing all openings in the surface, joints and seams of ductwork with adhesives, mastics, sealants or gaskets in accordance with AS 4254 for a Class C seal. Flexible ductwork must be sealed with a draw band in conjunction with a sealant or adhesive tape.

Under 3.12.5.3(b) the duct insulation located in the garage and roof space is required to be protected by an outer sleeve of protective sheeting which is sealed with adhesive tape of minimum 48 mm width.

8.4 Electric resistance space heating

The only electric resistance space heating is the in-slab heating system of the bathroom and it will have a thermostat and time switch. At 120 W/m², the power load is less than the 150 W/m² allowance.

8.5 Artificial lighting

As the house is 116 m² in area and the garage 18 m² in area, the allowances provided by Part 3.12.5.5 are 580 Watts and 54 Watts respectively.

The proposed lighting system has a lighting load of 500 Watts for the house and 60 Watts for the garage. Upon review of the allowances above, it can be determined that the house lighting complies but the garage lighting load will need to be reduced to not more than 54 Watts.

8.6 Heating and pumping of a swimming pool or spa pool

There is no swimming pool or spa pool so the provisions of Part 3.12.5.7 do not apply.

9. SUMMARY

This case study provides detailed analysis of the energy efficiency DtS Provisions of Volume Two and how they apply to a typical masonry veneer house proposed to be constructed in climate zone 6. The DtS Provisions are based on the eight climate zones that classify Australia into broad regional areas with similar climatic conditions.

For the building fabric, Part 3.12.1 sets out the levels of thermal efficiency for roof and ceiling constructions, roof lights, external walls, floors and the attached garage.

Part 3.12.2 must then be examined to establish the treatment of the external glazing to control the amount of heat entering or leaving the house through glazing. The glazing calculator developed by the ABCB assists with calculations required under this part of Volume Two.

To seal the building, Part 3.12.3 has DtS Provisions that address unwanted air leakage, which must be examined to verify sealing measures for chimneys and flues, roof lights, external windows and doors, exhaust fans and evaporative coolers and general construction requirements for roofs, walls and floors.

The next step is to examine the requirements of Part 3.12.4 relating to beneficial air movement within the building to determine the applicable provisions for air movement, ventilation openings and ceiling fans and evaporative coolers. As the location of the proposed building is climate zone 6 there are no additional provision over that required for health.

The last step is to determine the requirements for services in Part 3.12.5, namely heating and cooling services, lighting, and swimming pool or spa pool heating and pumping, to prevent excessive energy usage.